COST ESTIMATION AND BUDGETING

LECTURE 9

G.R. Labbé & R.W. Schott

Cost Management

- Cost Management starts during the planning phase of a project when a reasonable budget is determined. Identify project resources *(human and material)* and a time-phased budget reflecting their involvement.

Some of the more common sources for cost in a project are:

- Labor
- Materials
- Sub-contractors *(external or internal support departments like the test or drafting department)*
- Equipment and Facilities
- Travel
- Shipping

Cost Estimation

- Cost estimation is the first step in developing a comprehensive budget for the project and determining if the project is feasible or not.
- In order to develop this cost estimation it is critical to determine all the costs that might be associated with the project and categorize them into one or more of 8 types.
- One thing to remember is that a cost can belong to more then one category of cost.

Direct vs. Indirect Cost

- **Direct costs** are those clearly generated by the project.
- Labor and Materials are good examples of direct cost. For example, all PO's can be summarized and are allocated as a direct cost.
- The calculation of the total labor cost is straight forward:

Direct Labor Rate x Hours Spent = Total Direct Labor Cost

Direct vs. Indirect Cost

- **Indirect costs** are typically overhead cost.
- They come mostly from cost of sales, overhead, and general administration. Property tax, rent, health and benefit cost for the employees, depreciation of equipment, indirect materials, and utilities are some examples of indirect cost.
- Indirect cost is harder to identify, and many organizations choose a percentage approach. That means that the indirect cost are being expressed as percentage of direct cost. Usually in the 20% to 50% range.
- Other companies apply the indirect cost by a project-by-project base.
- Whichever is preferred by the company, all project estimates include both direct and indirect cost estimates.

Fixed vs. Variable Cost

- Fixed cost don't vary with their usage. So they are a fixed cost for the project.
- For example the cost of leasing certain pieces of equipment does not usually change with the amount of usage they get. They are leased for a certain period of time.
- This type of cost makes leasing expensive equipment a difficult decision for a Project Manager. They must decide if the amount of use the equipment will see justifies the costs.

Fixed vs. Variable Cost

- Variable costs are calculated as the cost proportional to the amount of usage the resource gets.
- Rental equipment is often charged a rate based upon the amount it is used. *(Rental Cars – Mileage)*. The more a piece of equipment is used the more it depreciates in value, therefore the more it costs to use it.
- It is common for projects to have a mix of costs that are both fixed and variable.
- **What other examples can you think of?**

Normal vs. Expedited Cost

- Normal cost refers to the cost incurred during normal operation, during normal working hours.
- These are costs are routine costs associated with completing the scheduled work agreed upon by the stakeholders at the start of the project.
- They can include overtime costs as long as the hours were already scheduled.

Normal vs. Expedited Cost

- Expedited cost is the cost that is required for extra fast or even dedicated shipments or hiring extra people to get a task done quicker.
- They are unplanned costs that are associated with steps taken to speed up the completion of a task or tasks to get back on schedule.

Recurring vs. Nonrecurring Cost

- **Nonrecurring cost**
- Are typically costs charged either at the beginning or end of a project and can include items such as personnel training or preliminary market analysis.
- It is usually viewed as a one-time expense.

Recurring vs. Nonrecurring Cost

- **Recurring cost**
- Is usually associated with logistics cost, utilities, labor costs and the like.
- These are expenses that will occur again during the life of the project.
- During the budget management and cost estimation process it is very important to highlight which are recurring and which are nonrecurring costs.
- This is especially true when a time-phase budget is being worked on.

Cost Estimation

Estimating the cost of a project, at the very beginning, is a very challenging task. There are two important project principles that should be followed every time this is done.

- First, the more clearly you define the various cost estimates upfront, the less likely it is to have budgeting errors.
- Second, the more accurate your initial cost estimate, the more likely it is to have a budget that reflects reality and the less likely it is to have budget overruns later.

Cost Estimation Methods

- **Ballpark Estimates**

Are used when the time or resources are very scarce to go through a thorough cost estimation process. It also can be the initial estimate to see if a company wants to spend even the resources on bidding for a certain contract. The accuracy is typically **±30%.**

- **Comparative or Parametric Estimates**

This is used when a company has experience with another project already that is close to the new one. Then the baseline project numbers can be taken and if carefully applied, an accurate estimate can be the result. Care must be taken that a solid history of comparable projects is available. This requires a reliable archiving system. Literature argues that this type of estimate is accurate to **± 15%**. However if done carefully, results can be much better.

Cost estimation methods *(cont'd...)*

- **Feasibility estimates**

These are calculated after the first quotes from suppliers come back, when the scope is clearly defined. Usually used in the construction industry where published tables of material cost exist, and the amount of material used is known at this time. The accuracy is about **±10%** because it is performed further down the development process.

- **Definitive estimates**

These can only be given after the completion of most of the design steps. The major purchase orders can be written, the detail is known and therefore the accuracy is the best. For this type of estimate an accuracy of **±5%** is accepted.

Which one should a company use?

That depends on the experience of the company and the team.

How many similar projects have been delivered?

What type of project is it?

For example, a R&D project for a new design will likely have a ± 20% range, whereas a banquet for 200 people, will have a relatively precise cost estimate that can be given upfront.

Factors Influencing the Quality of Estimates

- **Planning Horizon**
  Estimates of current events are close to 100 % accurate. Generally, the further the task is in the future, the less accurate it is.

- **Project Duration**
  New technology is especially challenging to estimate if the Scope statement is written poorly. Long duration projects increase the uncertainty in estimates.

Factors Influencing the Quality of Estimates *(cont'd...)*

- **People**
  The estimates provided depend on the skill of the people making it. The closer the experience is matched to the person, the more accurate the estimate. Also if the project team has worked together before will influence the time to form into an effective team. Staff turnover can influence estimates.
- **Project Structure and Organization**
  Dedicated project teams will deliver faster results, but it ties up personnel full time for the duration of the project. Projects in matrix organizations will run more cost efficient, but slower.

Factors Influencing the Quality of Estimates *(cont'd…)*

- **Padding Estimates**
  Usually estimates are padded for added safety. If everyone adds a little padding to the task, the project time and cost are severely overstated, and doesn't reflect the true reality.

Factors Influencing the Quality of Estimates *(cont'd...)*

- **Organization Culture**
  Some organizations encourage padding, others place a high priority on accurate estimates. Also depending on the consequences the estimator must face if the team member does not meet the estimate determines the estimate. A clear distinction between "Estimate" and "Commitment" must be made.
- **Other Factors**
  Non project related factors also have an impact on the estimation accuracy. Some factors are for example: equipment downtime, sickness, vacation, holidays, legal limits, project priority.

Creating A Project Budget

- ✓ The process of developing a project budget is an interesting mix of estimation, analysis, intuition, and repetitive work. The central goal of a budget is the need to support rather than conflict with the project's goals.
- ✓ Perhaps most importantly, the project budget and project schedule must be created in tandem. The budget effectively determines whether or not project milestones can be achieved.

Creating A Project Budget

- ✓ Several important issues go into the project budget, including cash flow - income and expenses *(tools, prototypes, labor,...)*
- ✓ Expected revenue streams including *(expected sales, royalties, ...)*

Top-Down Budgeting

- Requires the direct input from the organizations top-management.
- This approach interviews senior management to get their opinion and experiences regarding the cost estimated.
- This assumes that senior management is experienced with past projects and is in a position to provide accurate estimates for future project ventures.

Top-Down Budgeting *(cont'd…)*

- They take the first stab at estimating both the overall budget and the major work packages.
- These projections are then passed down the hierarchy to the next functional department level, where additional more detailed information is being collected.
- At each step down the hierarchy, the project is broken down into more detailed pieces, until project personnel who are actually doing the work provide the input on a task-by-task bases.

Disadvantages with Top-Down-Budgeting

- This approach can create a certain amount of friction within the organization.
- When top management establishes an overall budget at the start, they are in essence driving a stake in the ground saying: "This is all we will spend on this project!".
- As a result all levels below must make their estimates fit within the context of the overall budget.
- This process naturally leads to jockeying among different functions. The game has turned into a zero-sum-game. The more engineering gets, the less there is for purchasing to spend.

Advantages with Top-Down-Budgeting

- On the positive side, research suggests that top management estimates of project costs are often quite accurate, at least in the aggregate.
- Using this figure as a basis for drilling down to assign costs for work packages and individual tasks brings an important sense of budgetary discipline and cost control.

Bottom-Up Budgeting

- This approach begins inductively from the work break down structure to apply direct and indirect cost to project activities.
- The sum of the total costs associated with each activity are then aggregated, first to the work package level, then at the deliverable level.
- This goes up the WBS until the total project budget is calculated.

Bottom-Up Budgeting *(cont'd…)*

- Functional and senior management get involved at the respective levels of summarizing to ensure that no double counting is done and to find efficiencies with the rest of the organization.
- They are then responsible for creating the final master budget for the organization.

Advantages of Bottom-Up Budgeting

- This method emphasizes the need to create detailed project plans, particularly work break down structure.
- It also facilitates communication between project managers and functional managers.
- The unique creation of each project budget allows the top management a clear view for prioritization of scarce resources.

Disadvantages of Bottom-Up Budgeting

- It reduces the top managements control of the budget process to one of oversight, rather than direct initiation.
- This may lead to significant misalignments between the company's goals and the project goals.
- Also the fine-tuning often accompanying this process can be time consuming as top managers adjust to the new plan.

Activity-Based Costing (ABC)

- Activity Based Costing is a budgeting method that takes the belief that projects are based upon a series of activities and that each activity requires the use or consumption of a resource.
- Project activities are any discrete task that the project team undertakes to make or deliver the project.
- Therefore by calculating the cost of each activity you can determine the cost of the project.

Activity-Based Costing (ABC)

Activity Based Costing consists of four steps:

1) Identify the activities that consume resources and assign costs to them.
2) Identify the cost drivers associated with the activity. (Personnel, materials, ...)
3) Compute a cost rate per cost driver unit or transaction. Labor is commonly expressed as cost of labor per hour.
4) Assign costs by multiplying rates by units used by the project. E.g. an engineer's rate is $60/hr and she works for 80 hrs on the project. The total cost to the project would be $4800.

Activity-Based Costing (ABC)

- The table below demonstrates part of a budget. The purpose of this preliminary budget is to identify direct cost and those that apply to overhead expenses.

Activity	Direct Cost	Budget Overhead	Total Cost
Survey	3500	500	4000
Design	7000	1000	8000
Clear Site	3500	500	4000
Foundation	6750	750	7500
Framing	8000	2000	10000
Plumb and Wire	3750	1250	5000

Activity-Based Costing (ABC)

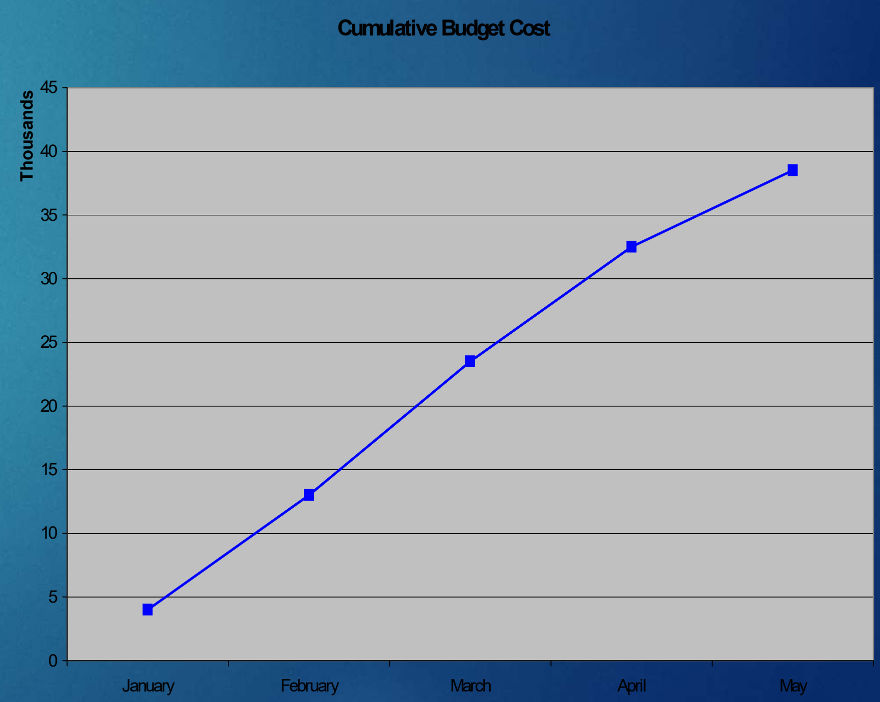
- The next table shows a budget in which the total planned expenses are compared against actual accrued expenses.

	Budget		
Activity	Planned	Actual	Variance
Survey	4000	4250	250
Design	8000	8000	0
Clear Site	4000	3500	-500
Foundation	7500	8500	1000
Framing	10000	11250	1250
Plumb and Wire	5000	5150	150
Total	38500	40650	2150

Time Phased Budgeting

- The next table shows the same budget in a time-phased budget. This time phased budget allocates costs across both project activities and the anticipated time in which the budget is to be expected.
- This allows the project budget baseline to be matched to the project schedule baseline identifying milestones for both schedule and budget.

Activity	January	February	March	April	May
Survey	4000				
Design		5000	3000		
Clear Site		4000			
Foundation			7500		
Framing				8000	2000
Plumb and Wire				1000	4000
Monthly Planned	4000	9000	10500	9000	6000
Cumulative	4000	13000	23500	32500	38500



Guidelines for Estimating *(Cost and Time)*

- **Responsibility**
  The person delivering the task should make the estimate *(except super-technical tasks)*. Draw on the experience! It will also ensure their "buy in" to the task.

- **Use several people to estimate**
  People with relevant knowledge will produce an agreed upon estimate. This ensures elimination of extreme estimates.

- **Normal conditions**
  Assumption for estimates should be normal working conditions, normal level of resources. They should always be defined.

Guidelines for Estimating *(Cost and Time)*

- **Time units**
  Estimation base needs to be defined.
- **Independence**
  Each task duration and cost should be estimated as it was independent of other items. Be careful when estimating sets of tasks one following the other. And the tasks need all to be estimated by the same person.
- **Contingencies**
  Work packages should **NOT** include allowances for contingencies. Top management needs to create an extra fund for contingencies that can be used to cover unforeseen events. Also the PM should have a project contingency fund he/she is responsible for.
- Adding risk assessment to the estimate helps to avoid surprises to the stakeholders.

Developing Budget Contingencies

- A budget contingency is the allocation of extra funds to cover uncertainties and improve chances that the project can be completed within the frame originally specified. Even the best budget is only an estimate and unknown circumstances will happen.

  - Contingency is typically added to the project's budget following the identification of all costs.
  - Contingency is not included in the budget generated with an Activity Based Costing.
  - Contingency is calculated as an extra cushion on top of the calculated cost.

Reasons why it makes sense to include contingency in project funding

- Project scope is subject to small adjustments.
  Not to confuse with changes. Changes are identified and tracked separately!
- Murphy's Law is always present
  It suggests that if something can go wrong, it often will.
- Cost estimation must anticipate iteration costs.
  For example a test was not successful because of an setup error and needs to be repeated.
- Normal conditions are rarely encountered
  One assumption for project planning is that the resources required will be available at the time when needed. This is not always the case!

Contingency in Project Funding

- Project teams naturally favor contingencies as a buffer for project cost control, however the acceptance by stakeholders and particularly clients is less enthusiastic.
- Some clients may object to the arbitrary process of calculating the contingency *(adding 10 to 15% to the total budget)*.
- Others may feel that they are being asked to cover poor budget control on the part of the vendor.
- Also there will be a discussion for which project activities contingency funds should be applied. Does the contingency apply across all tasks or should it be held for those tasks that are critical to the success of the project. These are decisions that need to be made early in the initial stages of the project and be incorporated into the scope statement.

Contingency in Project Funding

- Project cost estimation and budgeting are two important components of project management.

- The budget is such a significant constraint on any project, that it is best for the project team to prepare the estimates as careful as possible. Combined with a thorough risk analysis and a sufficiently detailed WBS this is the best defense against budget overruns.